Miguel Barcellano

Miguel.barcellano1379@gmail.com | linkedin.com/in/miguel-barcellano | https://miguel-barcellano1379.github.io/Portfolio/
Lewisville, TX | 972-567-7534

EDUCATION

The University of Texas at Dallas, Richardson, TX
Master of Science, Systems Engineering - Expected Spring 2027 | GPA: 3.4
Master of Science, Biomedical Engineering - Expected Spring 2027 | GPA: 3.5
Bachelor of Science, Biomedical Engineering - December 2025 | GPA: 3.3

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, C, HTML, CSS
Data & Machine Learning: Predictive Modeling, Data and Statistical Analysis, Signal Processing, Data Visualization
Engineering and Research Tools: SolidWorks, Arduino, Vicon Nexus, Visual3D, LabVIEW, GraphPad Prism
Testing and Instrumentation: SEM, XPS, Optical Microscope, Electrochemical Testing, Motion Capture Systems

PROFESSIONAL EXPERIENCE

BONE Lab, Richardson, TX - Graduate Research Assistant — April 2024 – Present

- Engineered and characterized surface-modified titanium implants using ionic-liquid coatings, SEM imaging, and electrochemical testing to improve biocompatibility and corrosion resistance for orthopedic
- Conducted biomaterials research focused on implant performance, osseointegration, and medical-device durability through surface characterization and experimental analysis

Neuromuscular Musculoskeletal Lab, Richardson, TX – Undergraduate Research Assistant — January 2022 – January 2024

- Captured and analyzed gait, motion-capture, and wearable EMG data using Vicon Nexus, Visual3D, and MATLAB to evaluate biomechanics, balance, and functional movement performance
- Supported biomechanics research through statistical analysis, study coordination, subject recruitment, and experimental data management focused on mobility and human movement

PROJECTS

Depression Severity Classification Using Oura Ring Sleep Data and Machine Learning, — January 2026 – May 2026

- Analyzed real-world wearable sleep and physiological biometrics collected from Oura Ring devices to investigate relationships between sleep behavior and depression severity
- Developed and optimized machine learning classification models using high-dimensional real-world wearable datasets, identifying Random Forest as the strongest-performing model for health prediction

Predictive Modeling of Suicide Risk Using Machine Learning Classification, — January 2026 – May 2026

- Developed machine learning classification models using real-world clinical and survey-based datasets to predict suicide risk in women with postpartum depression
- Performed data preprocessing, feature engineering, and model evaluation using AUROC, sensitivity, and specificity metrics, identifying neural networks as the highest-performing model for predictive accuracy

Effects of Cognitive Dual-Tasking on Biomechanics and Muscle Activity during Gait Initiation and Sit-to-Walk in Young Adults, — May 2023 – December 2024

- Analyzed gait biomechanics, wearable EMG, and motion-capture data using Vicon Nexus, Visual3D, and MATLAB to evaluate balance, mobility, and human movement performance
- Processed high-dimensional physiological and biomechanical datasets through statistical analysis techniques contributing to published research in PLOS One (2025)

Biomaterials Engineering and Surface Characterization of Titanium Dental Implants, — May 2025 – Present

- Engineered and characterized surface-modified titanium implants using ionic-liquid coatings, SEM imaging, and electrochemical testing to improve biocompatibility and corrosion resistance
- Conducted biomaterials research focused on implant performance, osseointegration, and medical-device durability through surface characterization and experimental